Eisenstein criterion Let $n \geq 1, f = a_0 + \ldots + a_n X^n \in \mathbb{Z}[X]$. If $\exists$ prime p s.t. $p \nmid a_n$, $p \mid a_0, \ldots, a_{n-1}, p^2 \nmid a_0$, then f is irreducible in $\mathbb{Q}[X]$.

Proof $p \nmid a_n \Longrightarrow p \nmid c(f)$, hence the same conditions hold for $\dfrac{f}{c(f)} \in \mathbb{Z}[X]$, i.e. may assume f is primitive.

If f is reducible in $\mathbb{Q}[X]$, then $f = gh, g = b_0 + \ldots + b_m X^m, h = c_0 + \ldots + c_l X^l \quad (m, l > 0)$, $g, h \in \mathbb{Z}[X]$ are both primitive. Apparently

$$\begin{aligned} a_n = b_m c_l &\Longrightarrow p \nmid b_m, p \nmid c_l, \\ a_0 = b_0 c_0 &\Longrightarrow \text{may assume } p \mid b_0, \end{aligned}$$

Take $1 \leq k \leq m$ s.t. $\begin{bmatrix} p \mid b_0, \ldots, b_{k-1} \\ p \nmid b_k \end{bmatrix}$, $a_k \equiv b_0 c_k + \ldots + \ldots + b_k c_0 \equiv b_k c_0 \equiv 0 \pmod p$.
From $p \nmid b_k$ we get $p \mid c_0 \Longrightarrow p^2 \mid a_0$. Contradiction! □

Examples

- $\forall n \geq 1, \exists$ irreducible $f \in \mathbb{Q}[X]$ with $\deg f = n$ by taking $f = p + X^n$ where p is a prime.
- Let p : prime, define $\Phi_p := 1 + X + \ldots + X^{p-1} = \dfrac{X^p - 1}{X - 1}$ is irreducible.

  Proof $\Phi_p = \dfrac{((X-1)+1)^p - 1}{X - 1} = \dfrac{\sum_{k=0}^{p} \binom{p}{k} (X-1)^k - 1}{X - 1} = \displaystyle\sum_{k=1}^{p} \binom{p}{k} (X-1)^{k-1}$.
  Consider the ring automorphism $\mathbb{Q}[X] \to \mathbb{Q}[X], f(X) \mapsto f(X-1)$ along with its inverse $g(X) \mapsto g(X+1)$, it preserves irreduciblity. Remains to show

  $$h(X-1) = \sum_{k=1}^{p} \binom{p}{k} (X-1)^{k-1}$$

  is irreducible.
  $h = \binom{p}{1} + \binom{p}{2} X + \ldots + \binom{p}{p-1} X^{p-2} + X^{p-1}$, with all non-leading coefficients $\in p\mathbb{Z}$.
  Eisenstein criterion $\Longrightarrow h$ irreducible in $\mathbb{Q}[X]$. □

  Remark Φ_p is the p-th cyclotomic polynomial (分圆/割圆多项式).
  In general, $\forall n > 1, \Phi_n = \displaystyle\prod_{0 \leq m < n,\, \gcd(m,n)=1} \left(X - e^{\frac{2\pi i m}{n}} \right) \in \mathbb{Z}[X]$ irreducible.

Remarks

- The general theory (including Eisenstein) works if $\mathbb{Z} \to R$: UFD and respectively $\mathbb{Q} \to K := \mathrm{Frac}(R)$. Can define $c(f)$, primitve polynomials, Gauss Lemma, etc.
- In particular R : UFD $\Longrightarrow R[X]$: UFD. Moreover $f \in R[X]$ is irreducible $\Longleftrightarrow$ either f is irreducible in R, or $\deg f > 0$, f primitive and f is irreducible in $K[X]$.
- R : UFD $\Longrightarrow \forall n \geq 1, R[X_1, \ldots, X_n]$ is UFD (recall: fields are UFD).
  $\because R[X_1, \ldots, X_n] \simeq (R[X_1, \ldots, X_{n-1}])[X_n]$ by merging coefficients, and use induction on n.
- R : UFD, $f = a_n X^n + \ldots + a_0$. If $f\left(\dfrac{u}{v}\right) = 0, \dfrac{u}{v} \in K = \mathrm{Frac}(R)$, then $v \mid a_n$ and $u \mid a_0$.

  Proof Same as the case $R = \mathbb{Z}$. □

  Remark Given $f \in \mathbb{Z}[X] \setminus \mathbb{Z}$: primitive, $\exists$ algorithm to test irreducibilty (Kronecker).

To test: $\forall k \geq 1$, whether $\exists g, h \in \mathbb{Z}[X]$ s.t. $f = gh, \deg g \leq k$.
Choose $x_0, \ldots, x_k \in \mathbb{Z}$ distinct and $\forall i, f(a_i) \neq 0$.

$$f = gh \implies \forall i, g(x_i) \mid f(x_i) \text{ in } \mathbb{Z}$$
$$\implies \text{finitely possibilities for } (y_i) = (g(x_i))_{i=0}^{k} \in \mathbb{Z}^{k+1}.$$

Given $(y_i)_{i=0}^{k} \in \mathbb{Z}^{k+1}, \exists! g \in \mathbb{Q}[X]$ s.t. $\deg g \leq k, g(x_i) = y_i, \forall i$. Now we can test whether $g \in \mathbb{Z}[X]$ and $g \mid f$.

In practice: hard to compute, need to use some modulo methods etc.

§ Groups

Motivation "Symmetries" in mathematics, physics, etc.

Symmetries	Structure in Question
Automorphism $V \simeq V$	Vector spaces over F : field
Permutations on X	Finite set X
Orthogonal transformation	IPS/$_{\mathbb{R}}$ with $\dim V < \infty$
Unitary transformation	IPS/$_{\mathbb{C}}$ with $\dim V < \infty$
Rotations in $\mathbb{R}^3$	$\mathbb{R}^3$: IPS/$_{\mathbb{R}}$ + orientation

These are certain bijections on sets + structures. Common features include "identity", multiplication & composition, inverse, etc.

$\exists$ other math objects sharing these features!

Examples
- $R^\times$ $(R : \text{ring})$,
- invertible $n \times n$ matrices,
- $q \in \mathbb{H} : N(q) = 1$.

Definition **Group** $:= \text{data}(G, \cdot)$ satisfying
- G : set $\neq \emptyset$,
- $\cdot : G \times G \to G$ (i.e. binary operator on G), called "multiplication" $(x, y) \mapsto x \cdot y$,
- $x(yz) = (xy)z, \forall x, y, z \in G$,
- $\exists 1_G \in G$ s.t. $x \cdot 1_G = x = 1_G \cdot x, \forall x \in G$,
- $\forall x \in G, \exists x^{-1} \in G$ s.t. $xx^{-1} = 1_G = x^{-1}x$ (we will show later that 1_G is unique for G).

Abbreviate $(G, \cdot)$ as G.

We say G is **commutative/Abelian** if $xy = yx$ for all $x, y \in G$.

Basic properties
- 1_G is unique: if $1_G, 1'_G$ have the required properties, then $1_G = 1_G 1'_G = 1'_G$. (often written as 1.)
- Cancellation laws:
  - $xy = xz \implies y = z, \because (x^{-1}x)y = x^{-1}(xy) = x^{-1}(xz) = (x^{-1}x)z$.
  - $yx = zx \implies y = z$ for exactly the same reason.
- $\forall x \in G$, the term x^{-1} is uniquely determined by x since $xx^{-1} = x(x^{-1})'$ gives $x^{-1} = (x^{-1})'$.
- $(xy)^{-1} = y^{-1}x^{-1}$ by checking directly.

We say G is **trivial** if $G = \{1_G\}$. **Order** of $G := |G|$.

Variants

- If we remove the axiom about x^{-1} $\Longrightarrow$ get **monoid** (幺半群), preserves the uniqueness of 1_G.
- If both $\exists x^{-1}$ and $\exists 1_G$ are removed $\Longrightarrow$ get **semigroup** (半群).
- In a semigroup, can define

$$x^n := \underbrace{x \cdot \ldots \cdot x}_{n \text{ copies}} \quad (n \geq 1).$$

- In a monoid, can define $x^0 := 1$.
- In a group,, can define $\forall n \in \mathbb{Z}_{<0}, x^n = \left(x^{|n|}\right)^{-1} = \left(x^{-1}\right)^{|n|}$.

In each case $x^{m+n} = x^m \cdot x^n, (x^m)^n = x^{mn}$ and $x^{-n} = \left(x^{-1}\right)^n = (x^n)^{-1}$ with proper constraints.

Definition Let G : group, $H \subset G$. If
- $1_G \in H$,
- $x, y \in H \Longrightarrow x \cdot y \in H$,
- $x \in H \Longrightarrow x^{-1} \in H$,

Then we say H is a **subgroup** of G, in which case $(H, \cdot)$ is also a group.

Boring examples:
- $\{1_G\}, G$ are subgroups of G.
- $\bigcap H_i$ is a subgroup where $H_i \subset G$: subgroup.
- Center: $Z_G := \{z \in G \mid \forall g \in G, zg = gz\}$.
  Can check: Z_G is a subgroup and G Abelian $\Longleftrightarrow Z_G = G$.